AI for Product Development

Activity 08: Run a Prototype Experiment

C840 • v1.0

Evaluate a synthetic agent prototype, classify failures and recommend one causal improvement.

1. Purpose and deliverable

Evaluate a synthetic agent prototype, classify failures and recommend one causal improvement.

Required output: Experiment ledger with before-after scores and decision

Acceptance: The change targets one causal layer, the frozen set is preserved, and regression risk is documented.

2. Folder inventory

brief.yaml — machine-readable scope and decision contract

prompt.md — bounded agent instruction

data/mock-data.csv and mock-context.json — synthetic evidence

solution/expected-output.json — trainer reference shape

scripts/verify_activity.py — deterministic completeness check

evidence/checklist.md — learner sign-off record

3. Detailed procedure

1. Read brief.yaml and restate the product decision, output and acceptance rule.

2. Inspect the CSV and JSON inputs; list source IDs, dates, contradictions and missing evidence.

3. Open prompt.md and select an approved agentic AI builder. Do not add live data or integrations.

4. Ask the agent to plan the workflow before producing the artifact. Check roles, tools, data and stop conditions.

5. Execute the activity-specific workflow: Run baseline → Score outputs → Classify failure → Change one layer → Re-run set → Decide.

6. Require citations to source IDs and label every inference or assumption.

7. Review the output at the human gate; compare it with the acceptance rule and reject unsupported claims.

8. Save the artifact and evidence inside a learner working copy of this folder.

9. Run python3 scripts/verify_activity.py and retain the JSON result.

10. Complete evidence/checklist.md and record PASS, REPAIR or STOP with a named owner.

4. Troubleshooting

Missing citations: require row-level source IDs and rerun only the unsupported section.

Unstable recommendation: freeze criteria, rerun the same input and perform sensitivity analysis.

Verifier failure: compare the missing file or field with brief.yaml, repair it and rerun.

Unsafe action proposed: stop, remove live/external access and return the decision to the human owner.

5. Verification

Pass condition: The change targets one causal layer, the frozen set is preserved, and regression risk is documented.

The deterministic verifier checks package completeness; the human reviewer checks evidence quality and decision fitness.